

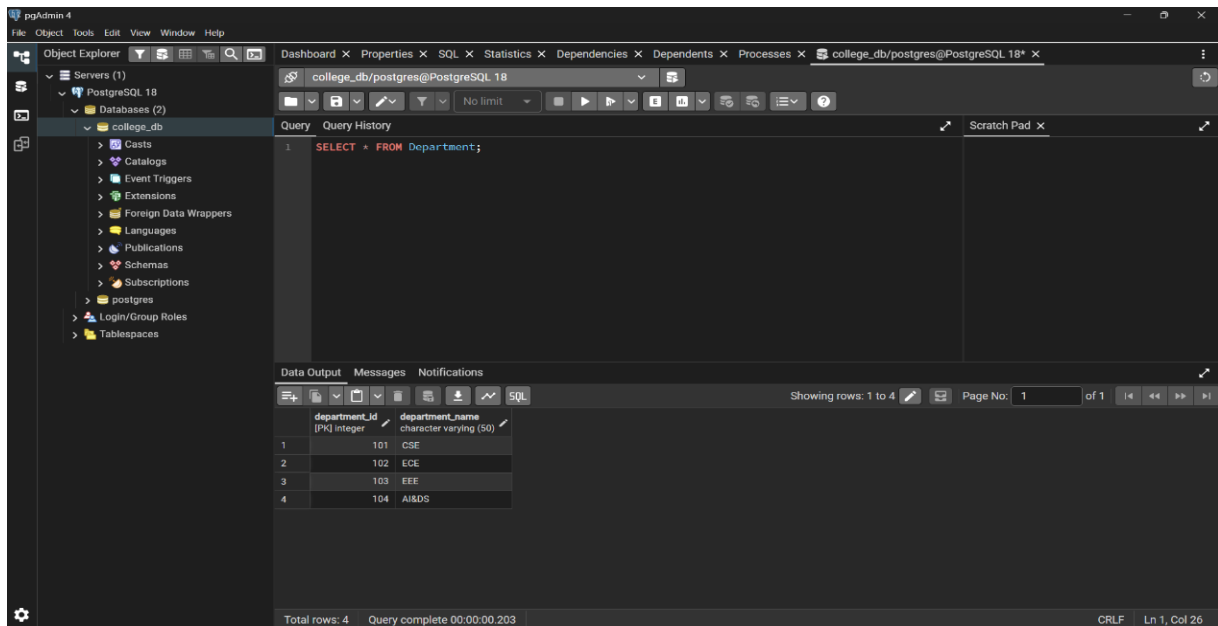
DSA0503 – Query Processing for Data Science

Activity using PostgreSQL

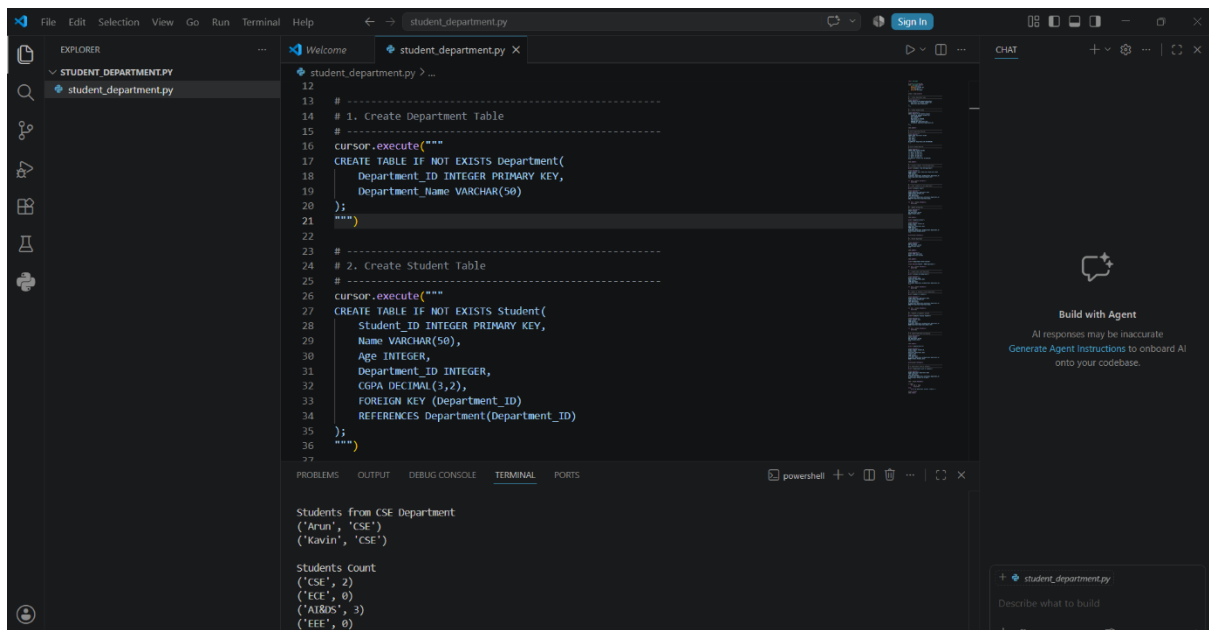
Divyesh S P (192424147)

1. Department Table (Parent Table) with attributes Department_ID

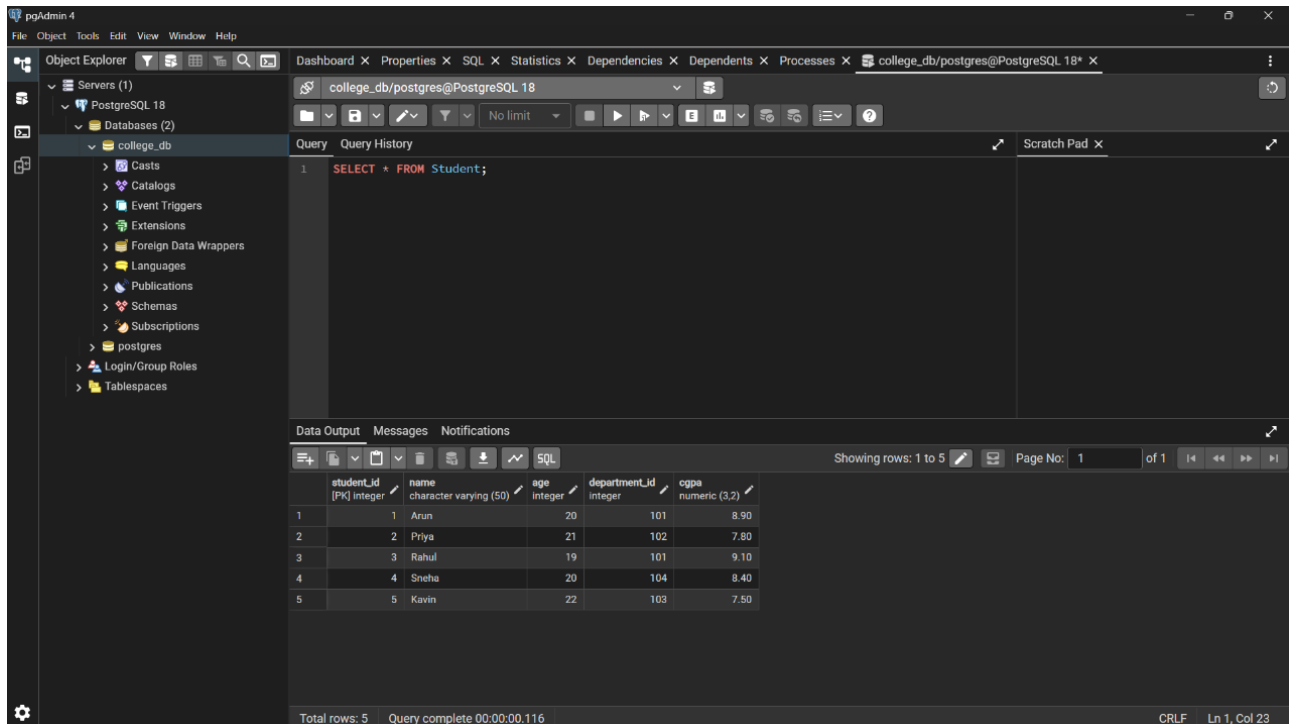
INTEGER PRIMARY KEY, Department_Name TEXT



Using Python:



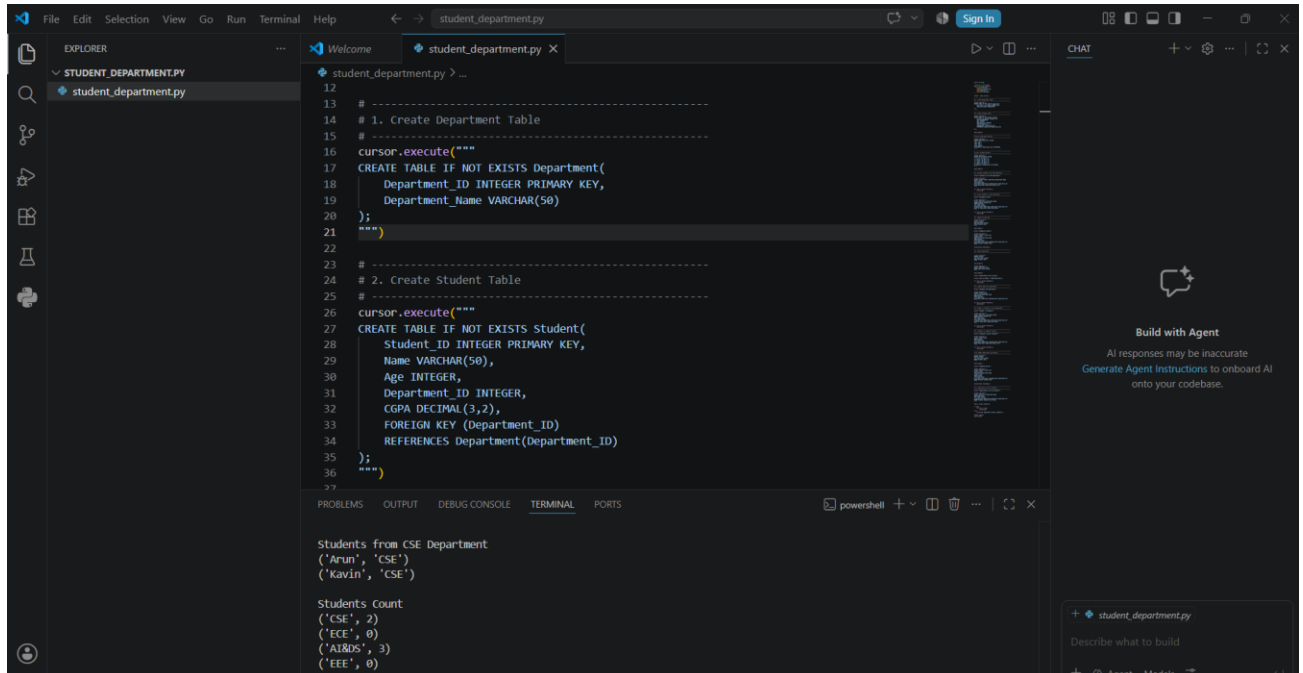
2. The Student Table (Child Table)



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, including the 'college_db' database and its tables. The main pane shows the 'Student' table with the following data:

student_id	name	age	department_id	cgpa
1	Arun	20	101	8.90
2	Priya	21	102	7.80
3	Rahul	19	101	9.10
4	Sneha	20	104	8.40
5	Kavin	22	103	7.50

Using Python:

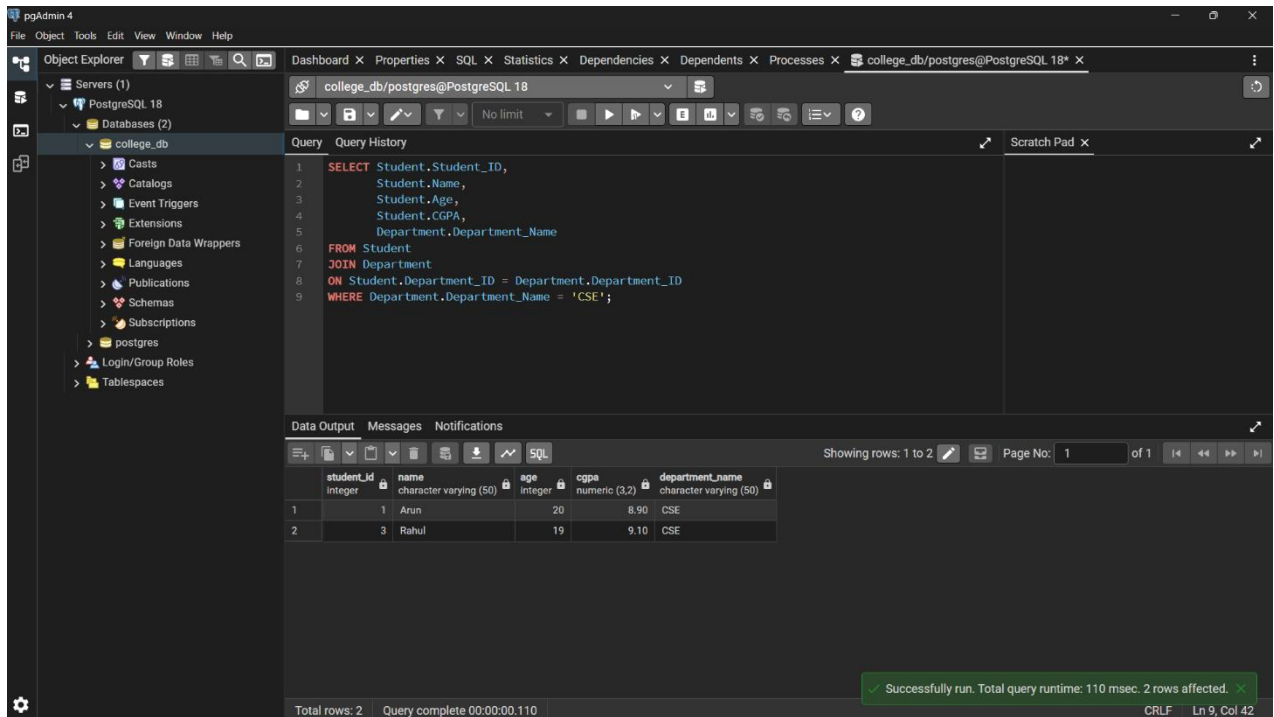


The screenshot shows a VS Code editor with a Python script named 'student_department.py'. The script creates a database schema and inserts data into the 'Student' table. The terminal output shows the results of the script execution:

```
Students from CSE Department
('Arun', 'CSE')
('Kavin', 'CSE')

Students Count
('CSE', 2)
('ECE', 0)
('AI&DS', 3)
('EEE', 0)
```

3. Displayed students from CSE department.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'college_db' under 'PostgreSQL 18'. The main pane shows a SQL query in the 'Query' tab:

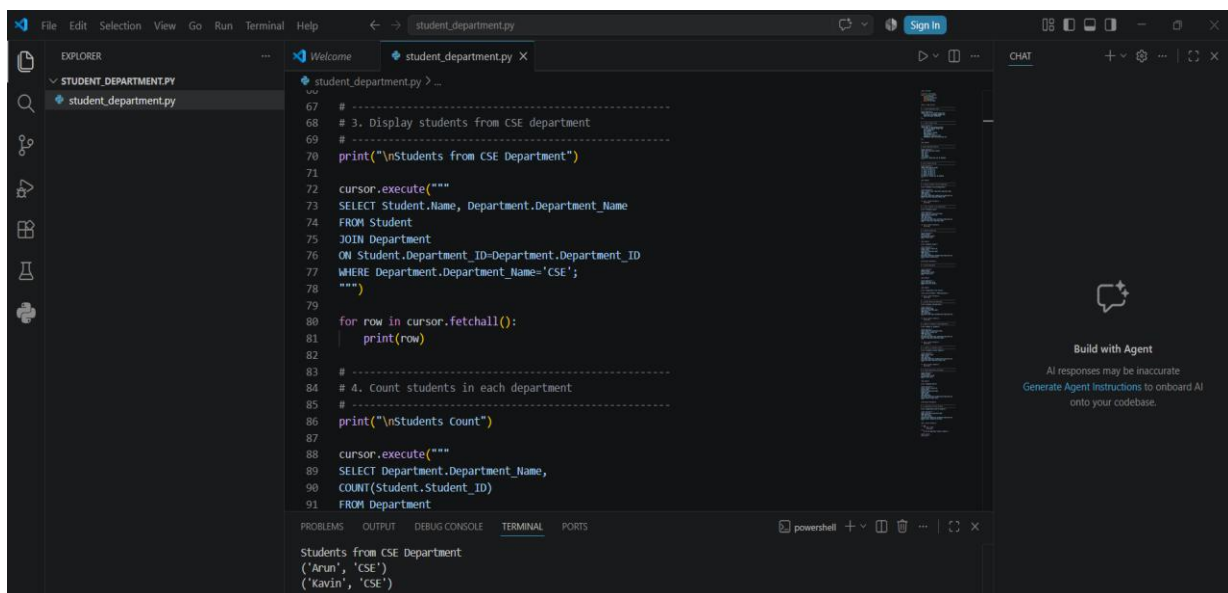
```
1 SELECT Student.Student_ID,  
2       Student.Name,  
3       Student.Age,  
4       Student.CGPA,  
5       Department.Department_Name  
6 FROM Student  
7 JOIN Department  
8 ON Student.Department_ID = Department.Department_ID  
9 WHERE Department.Department_Name = 'CSE';
```

The 'Data Output' tab shows the results of the query:

student_id	name	age	cgpa	department_name
1	Arun	20	8.90	CSE
2	Rahul	19	9.10	CSE

The status bar at the bottom indicates: 'Total rows: 2', 'Query complete 00:00:00.110', and a green message: 'Successfully run. Total query runtime: 110 msec. 2 rows affected.'

Using Python:



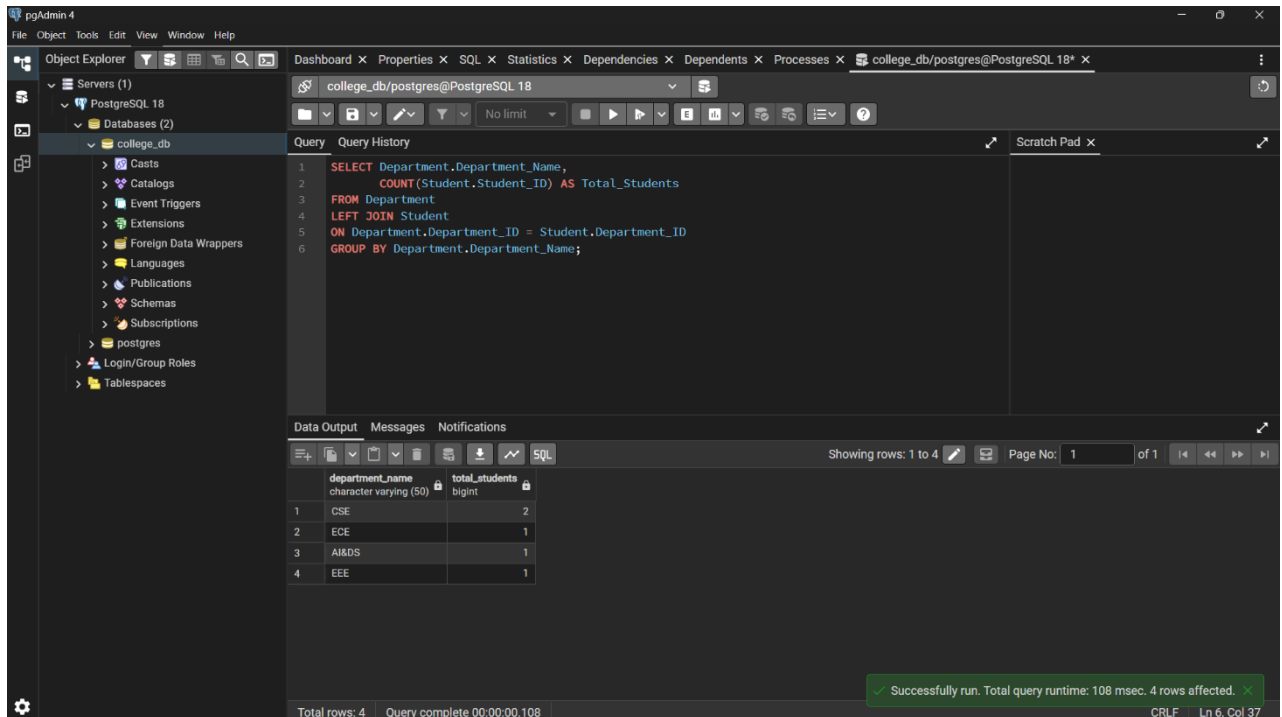
The screenshot shows a Python script named 'student_department.py' in a code editor. The script connects to a PostgreSQL database and executes two queries:

```
67 # -----  
68 # 3. Display students from CSE department  
69 # -----  
70 print("\nStudents from CSE Department")  
71  
72 cursor.execute("""  
73 SELECT Student.Name, Department.Department_Name  
74 FROM Student  
75 JOIN Department  
76 ON Student.Department_ID=Department.Department_ID  
77 WHERE Department.Department_Name='CSE';  
78 """)  
79  
80 for row in cursor.fetchall():  
81     print(row)  
82  
83 # -----  
84 # 4. Count students in each department  
85 # -----  
86 print("\nStudents Count")  
87  
88 cursor.execute("""  
89 SELECT Department.Department_Name,  
90 COUNT(Student.Student_ID)  
91 FROM Department
```

The terminal output shows the results of the first query:

```
Students from CSE Department  
(Arun, 'CSE')  
(Kavin, 'CSE')
```

4. Counted students in each department.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'college_db', including tables like 'Casts', 'Catalogs', 'Event Triggers', 'Extensions', 'Foreign Data Wrappers', 'Languages', 'Publications', 'Schemas', 'Subscriptions', 'postgres', 'Login/Group Roles', and 'Tablespaces'. The main pane shows a SQL query:

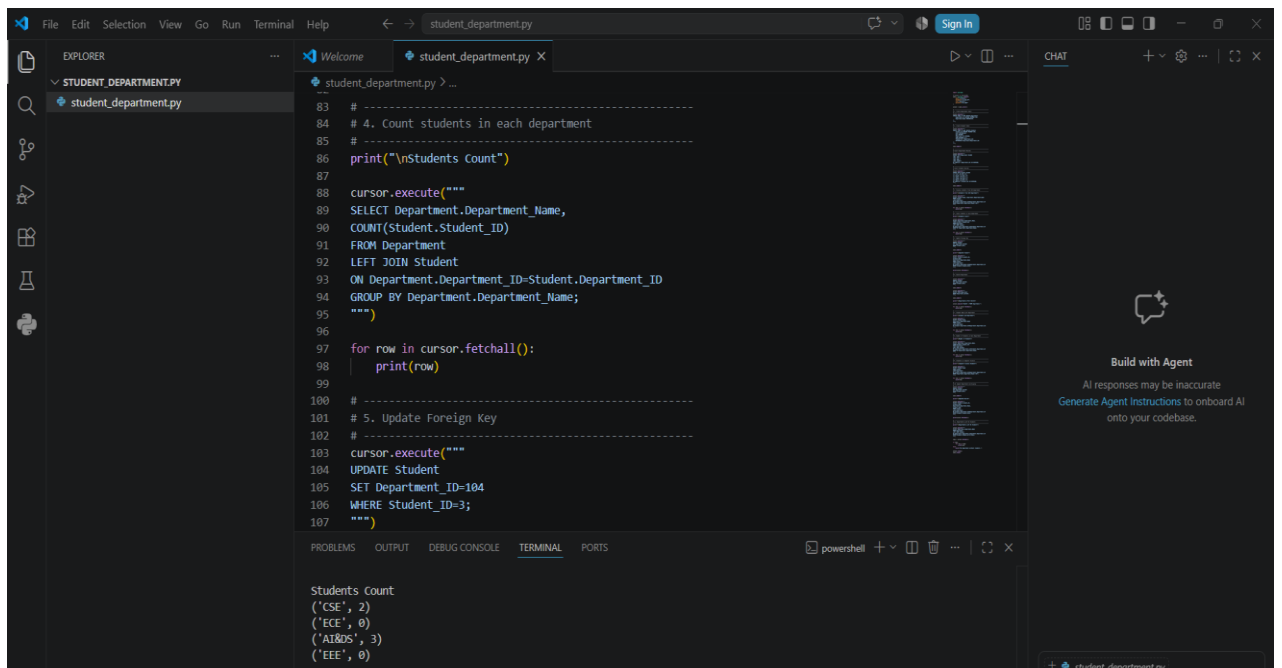
```
1 SELECT Department.Department_Name,  
2    COUNT(Student.Student_ID) AS Total_Students  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 GROUP BY Department.Department_Name;
```

The 'Data Output' tab shows the results of the query:

department_name	total_students
CSE	2
ECE	1
AI&DS	1
EEE	1

A status bar at the bottom indicates: 'Successfully run. Total query runtime: 108 msec. 4 rows affected.'

Using Python:



The screenshot shows a Python IDE with a file named 'student_department.py'. The script contains the following code:

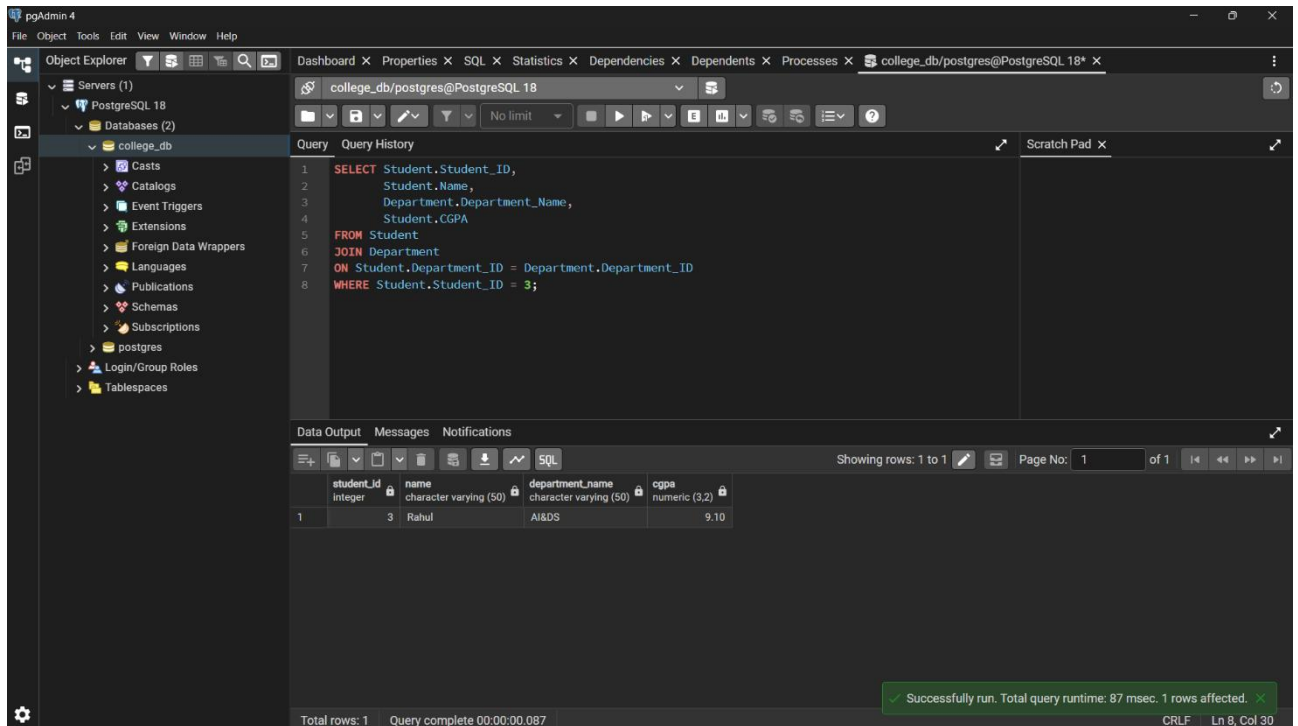
```
83 # -----  
84 # 4. Count students in each department  
85 # -----  
86 print("\nStudents Count")  
87  
88 cursor.execute("""  
89 SELECT Department.Department_Name,  
90 COUNT(Student.Student_ID)  
91 FROM Department  
92 LEFT JOIN Student  
93 ON Department.Department_ID=Student.Department_ID  
94 GROUP BY Department.Department_Name;  
95 """)  
96  
97 for row in cursor.fetchall():  
98     print(row)  
99  
100 # -----  
101 # 5. Update Foreign Key  
102 # -----  
103 cursor.execute("""  
104 UPDATE Student  
105 SET Department_ID=104  
106 WHERE Student_ID=3;  
107 """)
```

The 'TERMINAL' tab shows the output of the script:

```
Students Count  
( 'CSE', 2 )  
( 'ECE', 0 )  
( 'AI&DS', 3 )  
( 'EEE', 0 )
```

The right sidebar features a 'CHAT' panel with the text: 'Build with Agent. AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.'

5. Updated the foreign key.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'college_db', including tables like 'Casts', 'Catalogs', 'Event Triggers', 'Extensions', 'Foreign Data Wrappers', 'Languages', 'Publications', 'Schemas', 'Subscriptions', 'postgres', 'Login/Group Roles', and 'Tablespaces'. The main pane shows a SQL query being executed in the 'college_db/postgres@PostgreSQL 18' database. The query is:

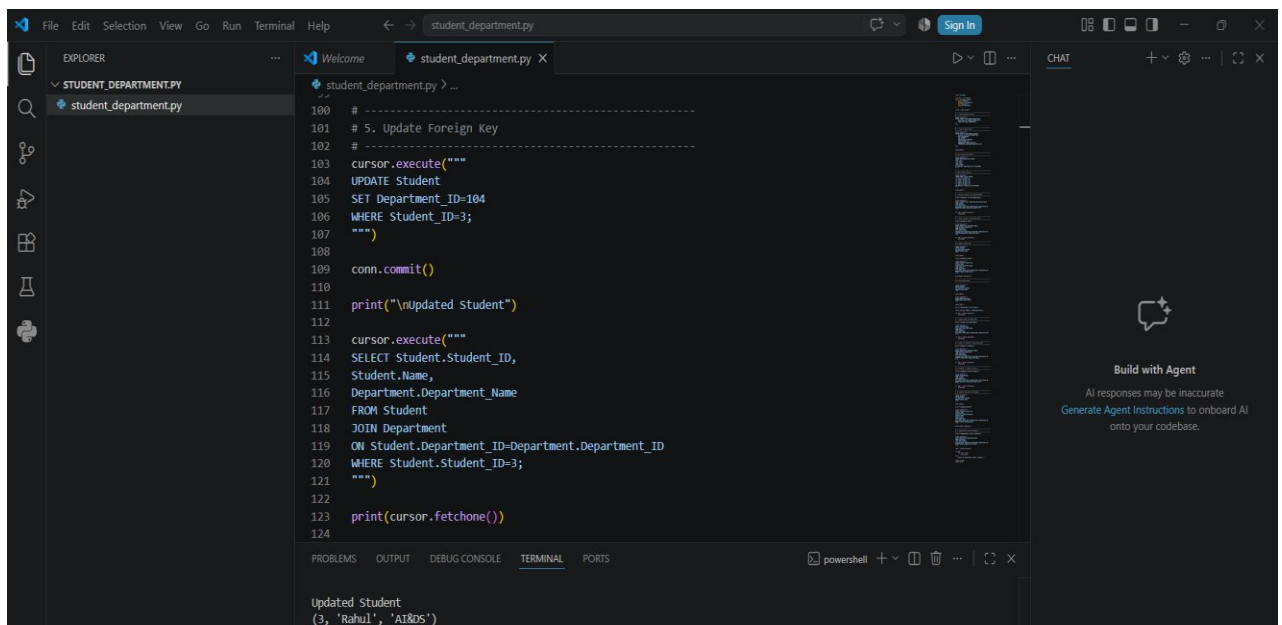
```
1 SELECT Student.Student_ID,  
2       Student.Name,  
3       Department.Department_Name,  
4       Student.CGPA  
5 FROM Student  
6 JOIN Department  
7 ON Student.Department_ID = Department.Department_ID  
8 WHERE Student.Student_ID = 3;
```

The 'Data Output' pane shows the results of the query:

student_id	name	department_name	cgpa
3	Rahul	AI&DS	9.10

A status bar at the bottom indicates: 'Total rows: 1 Query complete 00:00:00.087'. A green message box at the bottom right says: 'Successfully run. Total query runtime: 87 msec. 1 rows affected.'

Using Python:



The screenshot shows the VS Code interface with a Python script named 'student_department.py'. The script is used to update a foreign key in a PostgreSQL database. The code is as follows:

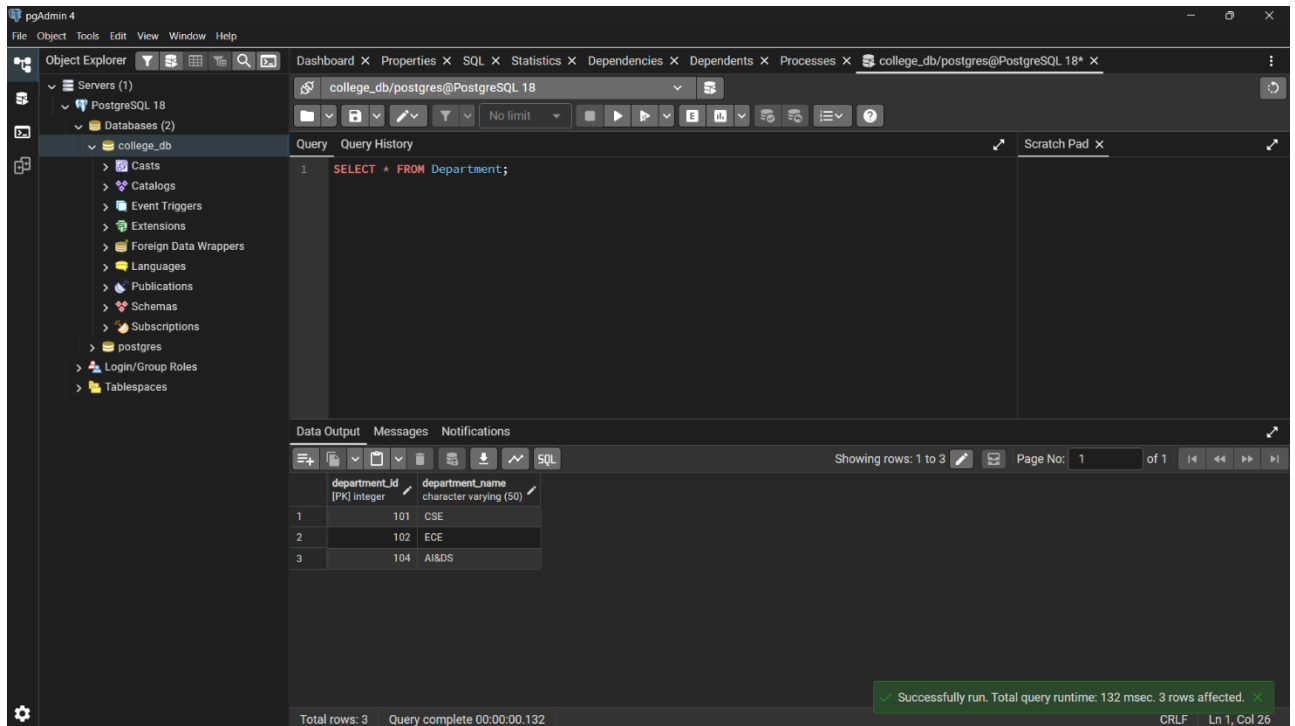
```
100 # -----  
101 # 5. Update Foreign Key  
102 # -----  
103 cursor.execute("""  
104 UPDATE Student  
105 SET Department_ID=104  
106 WHERE Student_ID=3;  
107 """)  
108  
109 conn.commit()  
110  
111 print("\nUpdated Student")  
112  
113 cursor.execute("""  
114 SELECT Student.Student_ID,  
115        Student.Name,  
116        Department.Department_Name  
117 FROM Student  
118 JOIN Department  
119 ON Student.Department_ID=Department.Department_ID  
120 WHERE Student.Student_ID=3;  
121 """)  
122  
123 print(cursor.fetchone())  
124
```

The output of the script is shown in the terminal:

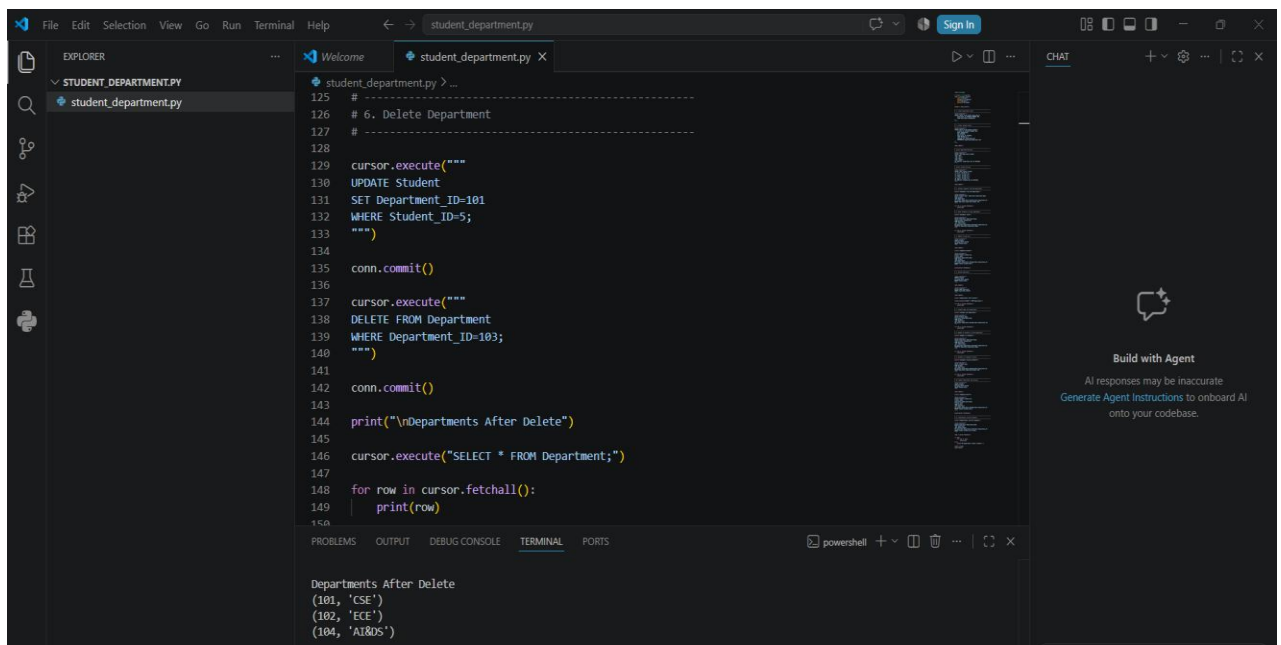
```
Updated Student  
(3, 'Rahul', 'AI&DS')
```

The right sidebar shows the 'CHAT' pane with a 'Build with Agent' button and a message: 'AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.'

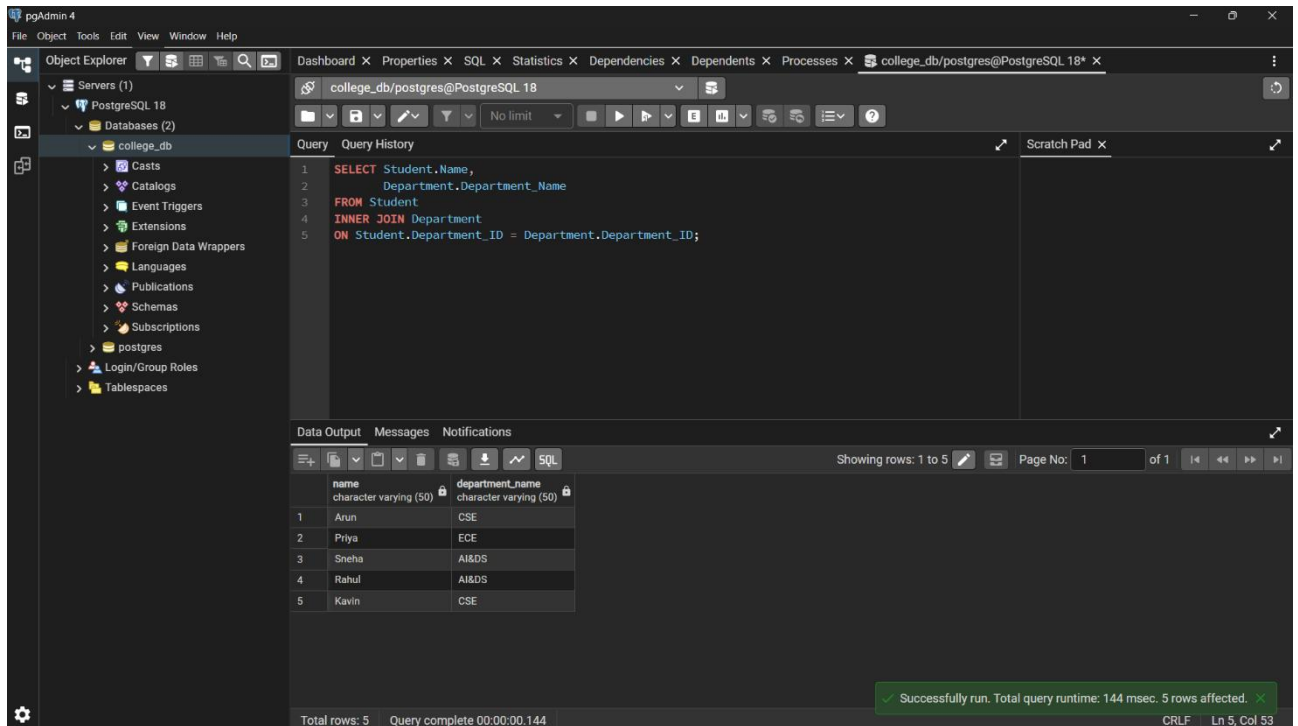
6. Deleted a department



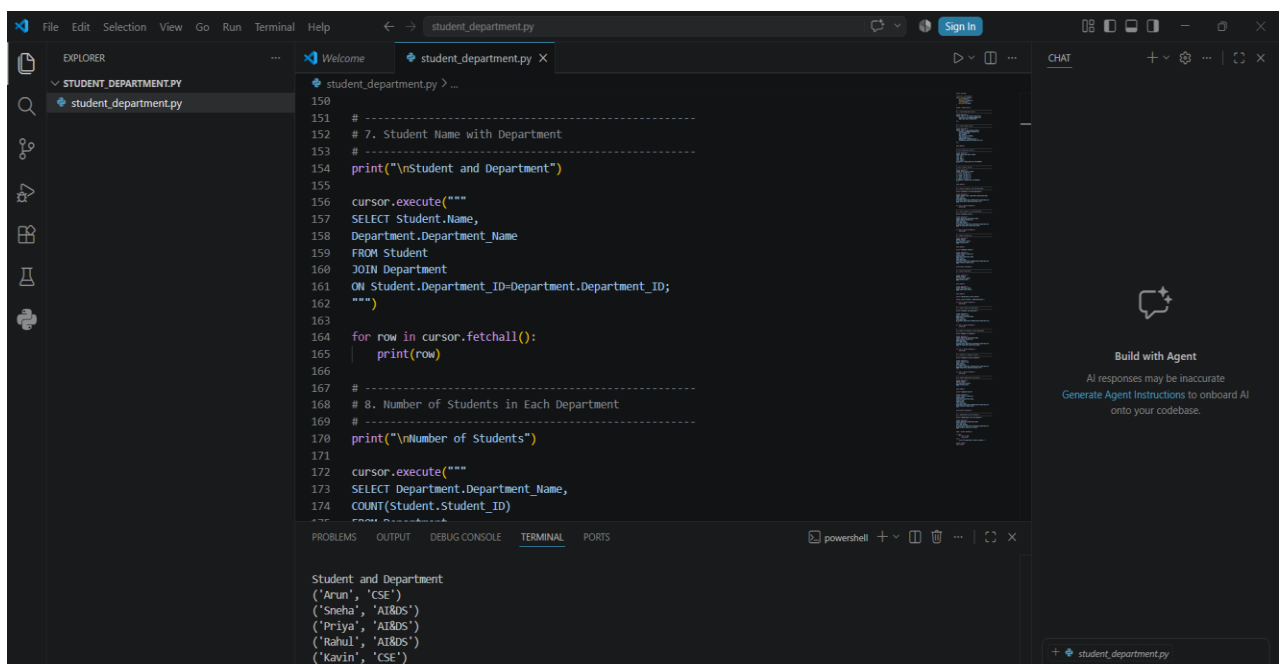
Using Python:



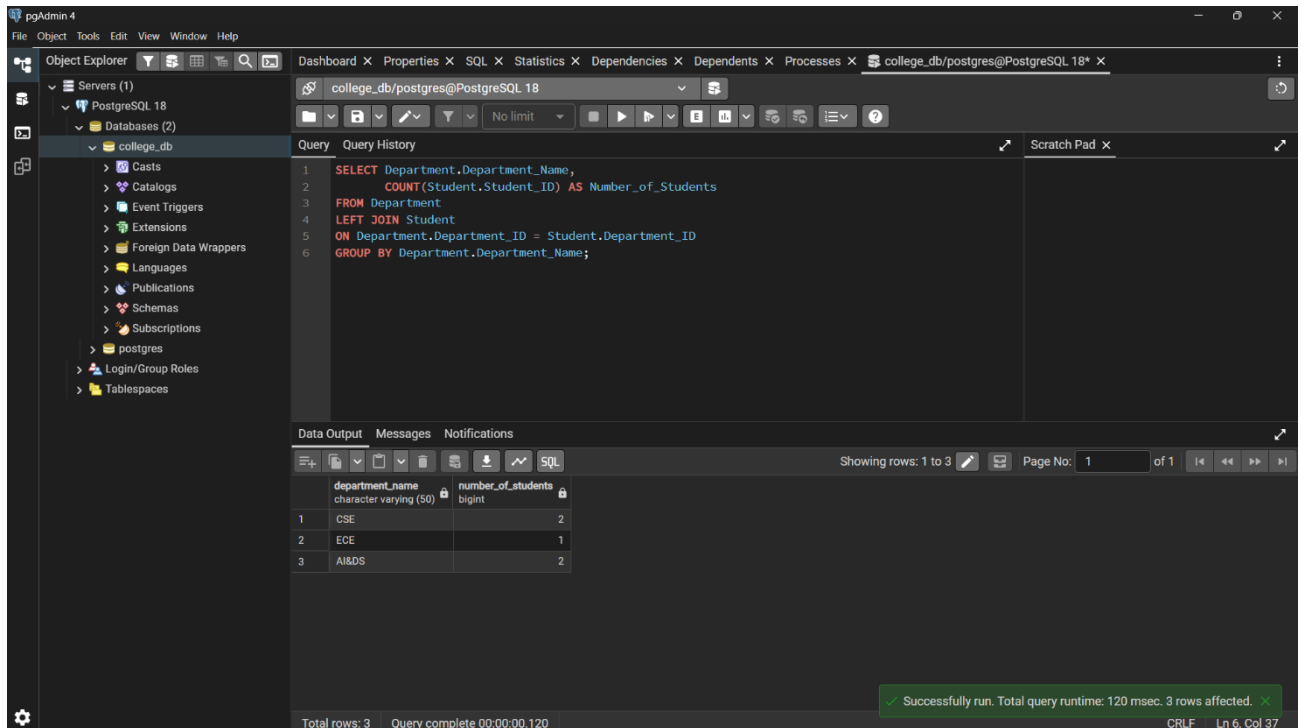
7. Displayed each student's name along with the corresponding department name using an SQL JOIN.



Using Python



8. Displayed the number of students in each department.



The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure for 'college_db', including tables like 'Casts', 'Catalogs', 'Event Triggers', 'Extensions', 'Foreign Data Wrappers', 'Languages', 'Publications', 'Schemas', 'Subscriptions', 'postgres', 'Login/Group Roles', and 'Tablespaces'. The main pane shows a SQL query:

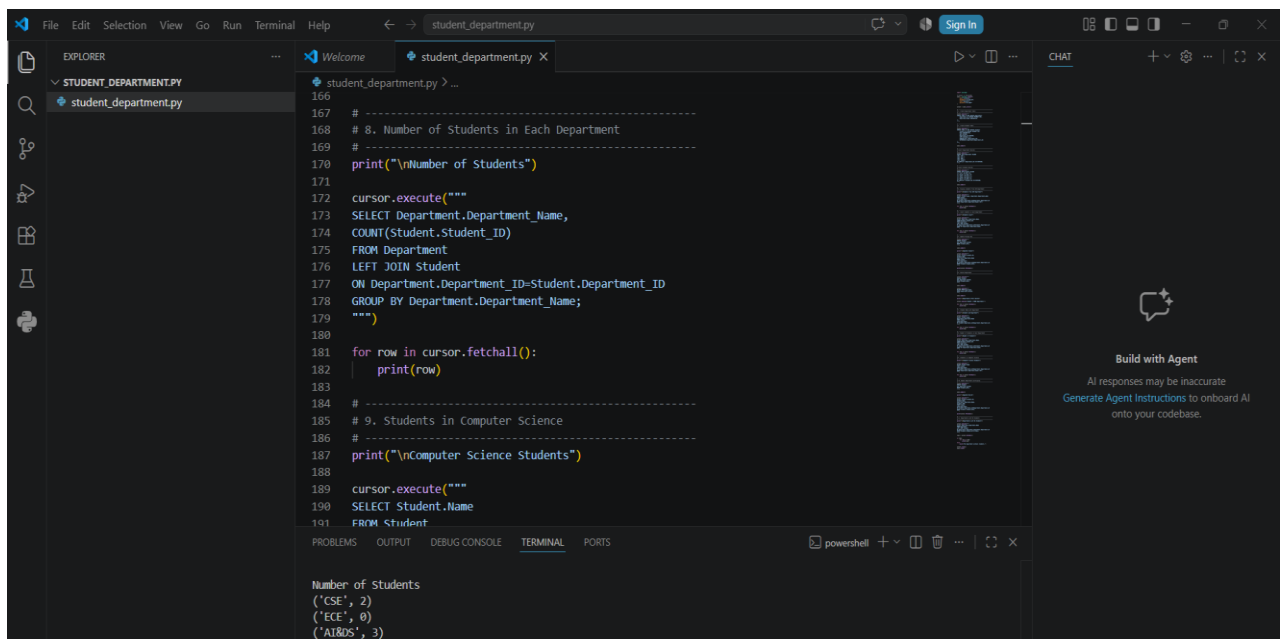
```
1 SELECT Department.Department_Name,  
2     COUNT(Student.Student_ID) AS Number_of_Students  
3 FROM Department  
4 LEFT JOIN Student  
5 ON Department.Department_ID = Student.Department_ID  
6 GROUP BY Department.Department_Name;
```

Below the query, the 'Data Output' tab shows the results in a table:

department_name	number_of_students
CSE	2
ECE	1
AI&DS	2

The status bar at the bottom indicates 'Total rows: 3' and 'Query complete 00:00:00.120'. A green message box at the bottom right states: 'Successfully run. Total query runtime: 120 msec. 3 rows affected.'

Using Python:



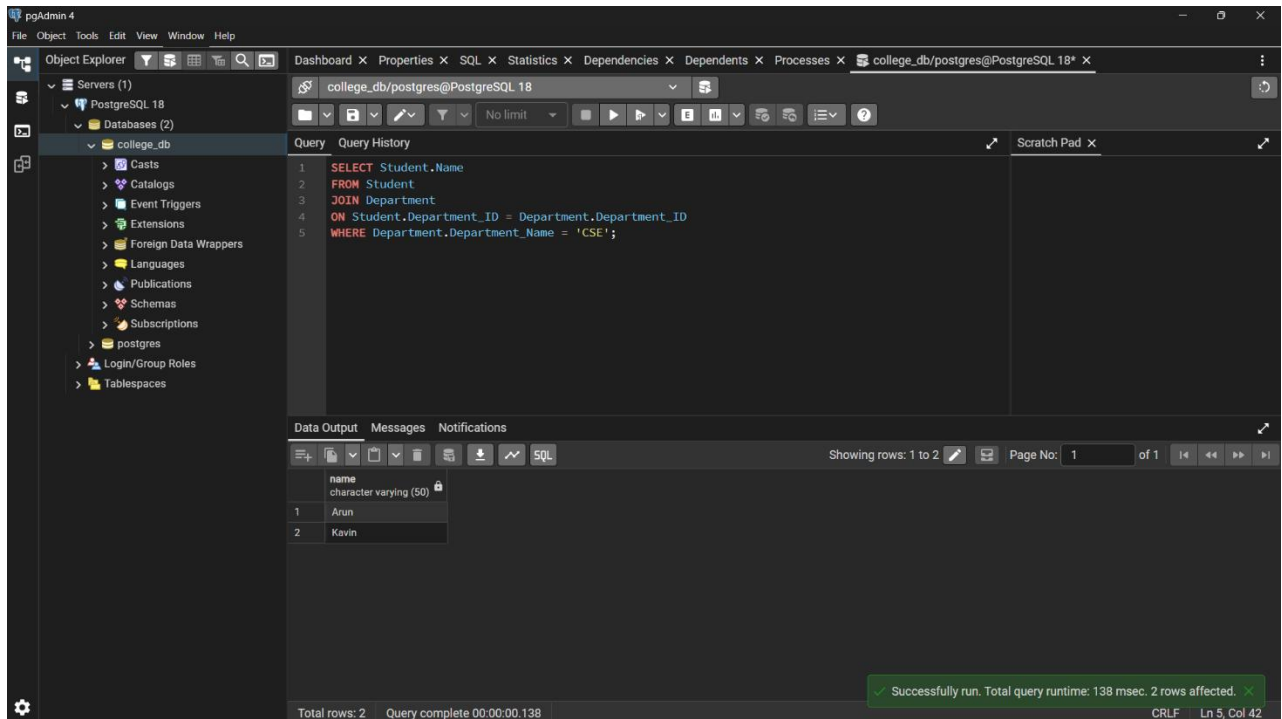
The screenshot shows a Python script named 'student_department.py' in a code editor. The script connects to a PostgreSQL database and executes a SQL query to retrieve the number of students in each department. The output is printed to the console.

```
166  
167 # -----  
168 # 8. Number of Students in Each Department  
169 # -----  
170 print("\nNumber of Students")  
171  
172 cursor.execute("""  
173 SELECT Department.Department_Name,  
174 COUNT(Student.Student_ID)  
175 FROM Department  
176 LEFT JOIN Student  
177 ON Department.Department_ID=Student.Department_ID  
178 GROUP BY Department.Department_Name;  
179 """)  
180  
181 for row in cursor.fetchall():  
182     print(row)  
183  
184 # -----  
185 # 9. Students in Computer Science  
186 # -----  
187 print("\nComputer Science Students")  
188  
189 cursor.execute("""  
190 SELECT Student.Name  
191 FROM Student
```

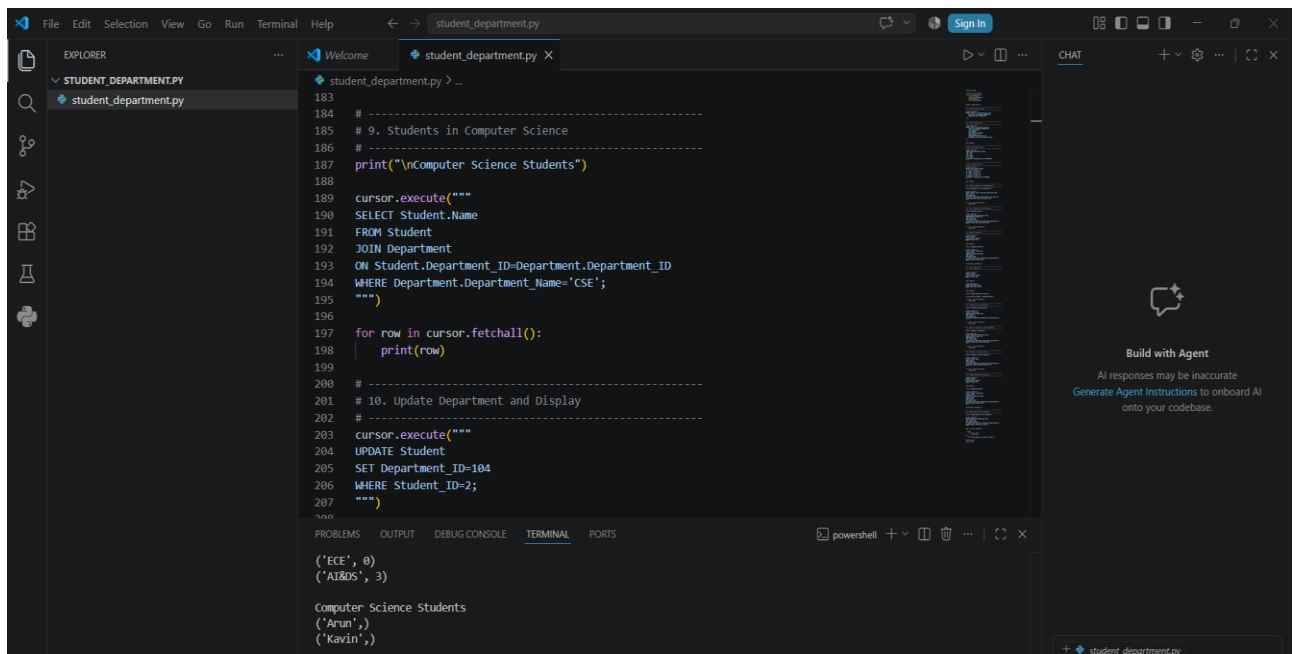
The terminal output shows the results of the query:

```
Number of Students  
(('CSE', 2)  
(('ECE', 1)  
(('AI&DS', 2)
```

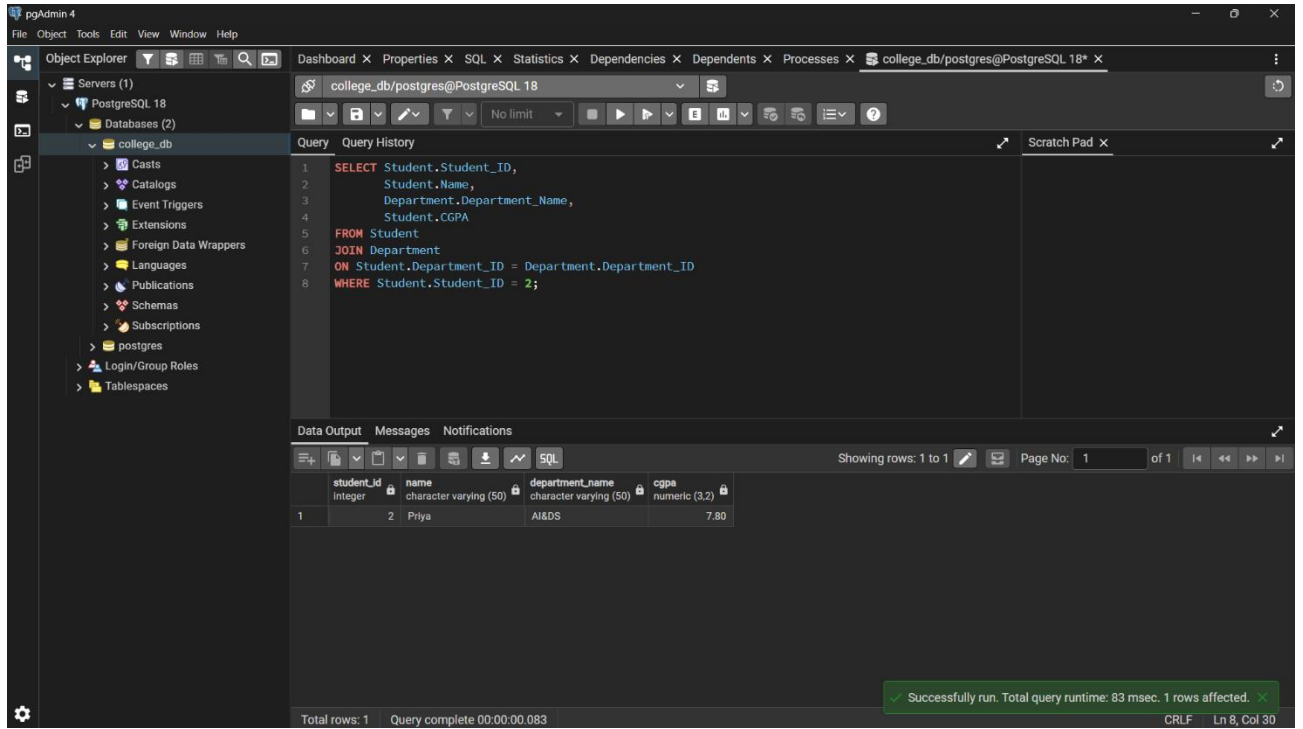

9. Displayed the names of students belonging to the Computer Science department.



Using Python:



10. Update the department of a student by changing the foreign key value and display the updated record.



The screenshot shows the pgAdmin 4 interface. The left pane displays the database structure for 'college_db', including tables like 'Students' and 'Departments'. The central pane shows a SQL query that updates the 'Department_ID' of a student with 'Student_ID' 2 to 104, and then displays the updated record. The query is as follows:

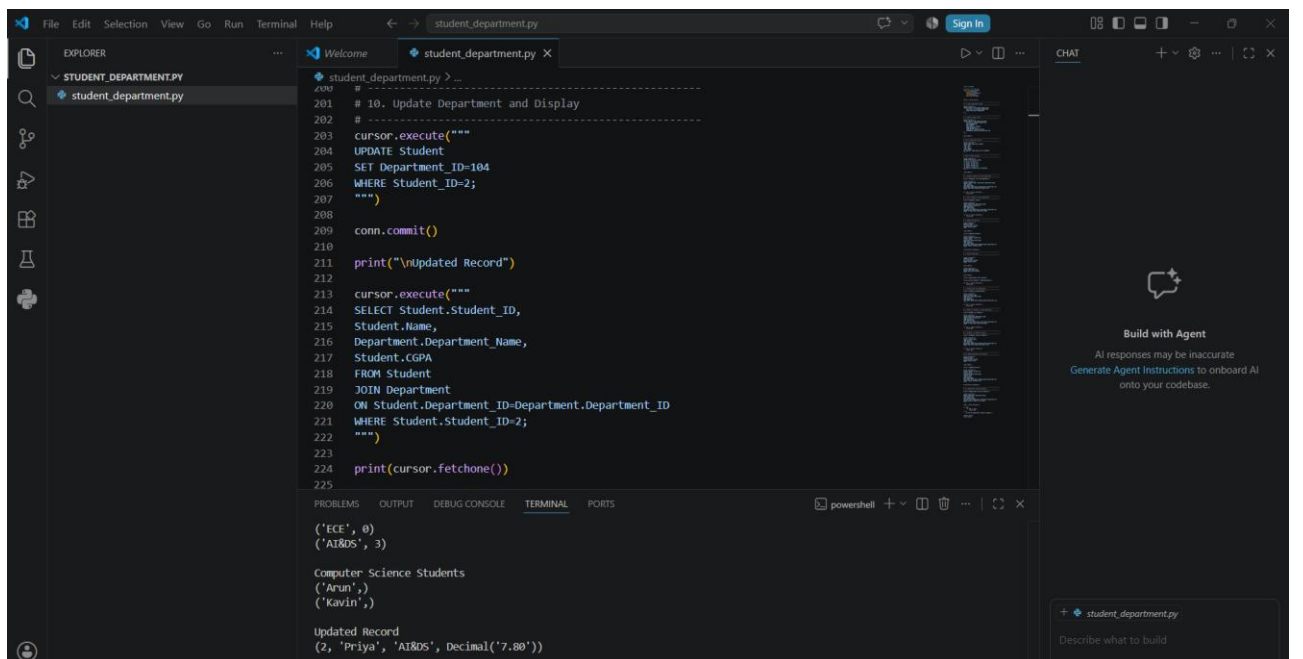
```
1 SELECT Student.Student_ID,  
2       Student.Name,  
3       Department.Department_Name,  
4       Student.CGPA  
5 FROM Student  
6 JOIN Department  
7 ON Student.Department_ID = Department.Department_ID  
8 WHERE Student.Student_ID = 2;
```

The bottom pane shows the result of the query, displaying a single row for the student with 'Student_ID' 2, named 'Priya', who is in the 'AI&DS' department with a 'CGPA' of 7.80.

student_id	name	department_name	cgpa
2	Priya	AI&DS	7.80

A status bar at the bottom indicates: 'Successfully run. Total query runtime: 83 msec. 1 rows affected.'

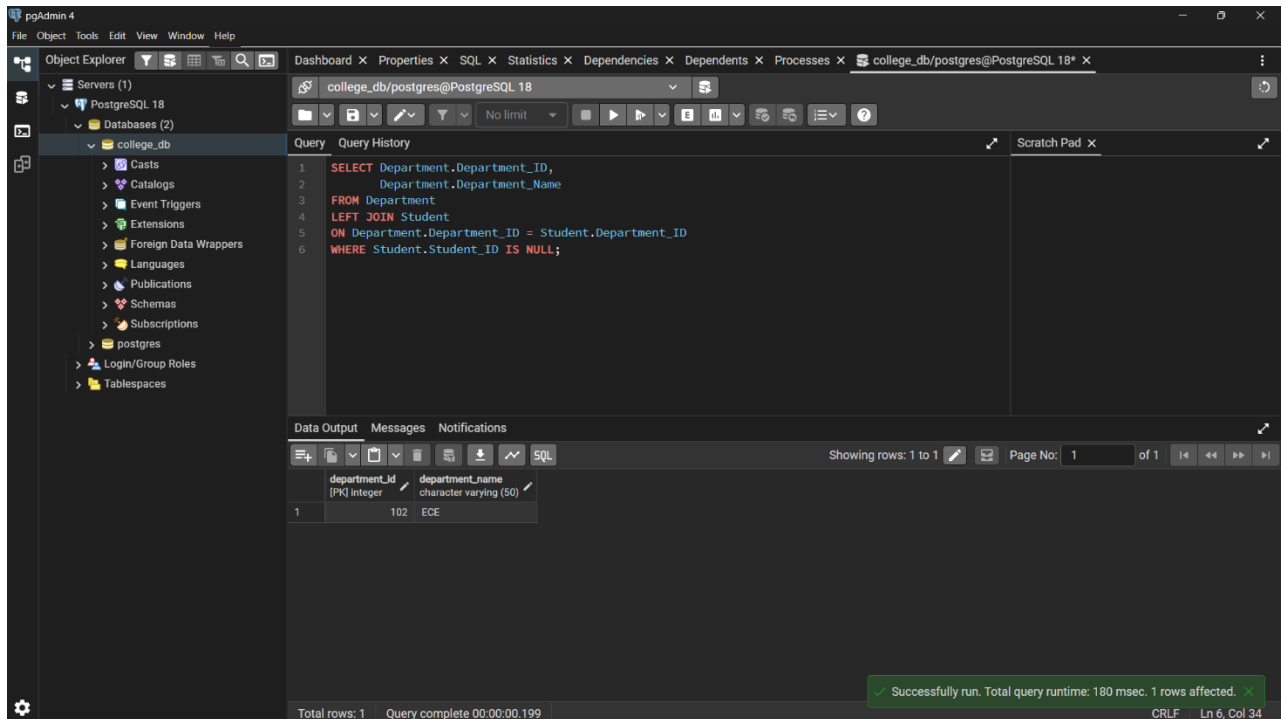
Using Python:



The screenshot shows a Python script named 'student_department.py' in a code editor. The script uses the 'psycopg2' library to connect to a PostgreSQL database, update the 'Department_ID' of a student with 'Student_ID' 2 to 104, and then display the updated record. The script is as follows:

```
1 # 10. Update Department and Display  
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11. Display departments that currently have no students assigned using an appropriate SQL JOIN.



Using Python:

